

Vigneshwar S Balakrishnan

Technical Architect · Principal Information Systems Engineer

12+ years architecting high-scale full-stack solutions across fintech & enterprise — scalable distributed microservices, multi-tenant platforms, white-label products, and developer experiences that power millions of users.

12+ YEARS EXP. 5M+ LOANS GOVERNED 40+ MICROSERVICES MIGRATED

Dallas, TX → Relocating to Chennai, India
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LANGUAGES

JavaScript TypeScript HTML5
CSS3

FRONTEND

React.js Next.js Gatsby.js
Redux Context API
Tailwind CSS

BACKEND

Node.js Express.js Seneca.js
Spring Boot REST APIs

CLOUD & DEVOPS

Azure AKS GCP Docker
Kubernetes Jenkins

DATABASES & MESSAGING

MongoDB MySQL Redis
Kafka RabbitMQ Socket.IO

API & SECURITY

Apigee OAuth 2.0 API Mgmt
JWT MFA

ARCHITECTURE

Microservices
Scalable Microservices
Event-Driven
Distributed Systems
Multi-Tenant Systems
White-Label Products
Cloud Migration Design Patterns

AI & PRODUCTIVITY

Claude Code GitHub Copilot

PRACTICES

Agile / Scrum Team Leadership
Tech Mentoring Code Review

CERTIFICATIONS

MERN Stack — NIIT Stackroute
Prompt Engineering & OpenAI — Columbia University
AWS Solutions Architect (In Progress)

EXPERIENCE

Technical Architect · Principal Information Systems Engineer

Jul 2018 – Present

Rocket Companies (Formerly Mr. Cooper)

Dallas, Texas, USA → Relocating to Chennai, India

Progressed through 5 levels: Software Engineer I → Software Engineer II → Senior Software Engineer → Lead Engineer → Principal Architect

- Architected zero-downtime cloud migration using Strangler Fig pattern — phased 40 interdependent microservices and on-prem DB servers to Azure/GCP, designing rollback safeguards, ADRs, and cross-team execution playbooks.
- Designed enterprise-scale OAuth 2.0 API Management gateway via Apigee — reduced API latency by 80%, governed API security across 5M+ loans, and established org-wide authentication standards adopted across all platforms.
- Architected event-driven HELOC and Mortgage Payment solutions — implemented Outbox Pattern for guaranteed at-least-once delivery and Kafka-based event streaming pipelines for real-time payment processing at scale.
- Led Rocket Mortgage platform integration — API contracts, data mapping, and Saga pattern orchestration workflows to coordinate distributed transactions between two of the largest US mortgage platforms.
- Architected multi-brand white-label mortgage servicing platform on a shared multi-tenant architecture; delivered Forbearance, Payments, Digital Home Advisor, and One-Time Links as Technical Lead using CQRS and Circuit Breaker patterns.
- Defined reference architectures, led architecture review governance, and authored ADRs adopted across 5+ product teams — establishing engineering standards for microservices, event-driven systems, and cloud-native design.

React Node.js Spring Boot Apigee OAuth 2.0 Kafka Azure Kubernetes Docker GCP Outbox Pattern Saga Pattern CQRS Strangler Fig

Analyst — Full Stack Engineer

May 2017 – Jun 2018

Verizon Data Services India

Chennai, India

- Built the Integrations Operations Portal — a real-time network monitoring single pane of glass for network engineers, with interactive UI and live event scheduling via Socket.IO & RabbitMQ.

React Node.js Socket.IO RabbitMQ Express.js

Project Engineer — Full Stack Developer

Sep 2014 – Apr 2017

Wipro Digital

Bangalore, India

- Engineered a reusable React component library for Lloyds Banking Group (UK's largest banks); full-stack developer on Quizart, an internal BFSI LOB application.

React Node.js Material UI Bootstrap Socket.IO

ARCHITECTURE HIGHLIGHTS

Zero-Downtime Cloud Migration

Rocket Companies · Principal Architect

Sole architect & execution lead for migrating on-prem DB servers and 40 scalable, distributed microservices to the cloud with zero downtime — phased strategy, rollback safeguards, cross-team coordination.

40 microservices · Full cloud adoption

Forbearance Application

Rocket Companies

Pandemic-era loan relief platform enabling customers to pause payments and enter modification workflows during COVID-19. Led complex UI modules and distributed service integration.

100K+ customers served

Event-Driven Payment Architecture

Rocket Companies · Principal Architect

Designed event-driven HELOC and Mortgage payment solutions using Outbox Pattern for guaranteed message delivery and Kafka event streaming for real-time processing. Eliminated dual-write inconsistencies across distributed payment services.

Guaranteed message delivery · Real-time payment at scale

Kafka Outbox Pattern Event Streaming Distributed Systems

OAuth APIM Gateway — Apigee

Rocket Companies · Principal Architect

Designed & implemented a large-scale OAuth 2.0 API Management gateway via Apigee, governing API security across the entire loan servicing ecosystem.

80% API latency reduction · 5M+ loans governed · Enterprise-wide API security

Rocket Mortgage Integration

Rocket Companies · Principal Architect

Architected integration layer — API contracts, data mapping, and Saga-pattern orchestration workflows for distributed transaction coordination between two of the largest US mortgage platforms.

Cross-platform integration at mortgage industry scale

Saga Pattern Spring Boot REST APIs Node.js

White-Label Servicing Products

Rocket Companies · Principal Architect

Architected white-label mortgage servicing products on a shared multi-tenant platform, enabling branded digital experiences for partner servicers at scale.

Multiple brand launches · Single multi-tenant platform

EDUCATION

Bachelor of Engineering — Electronics & Communications

2010 – 2014

Sri Krishna College of Technology · Coimbatore, India